

BIKASH TALUKDER

Full-Stack Developer & AI Systems Builder

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[GitHub](#) | [Portfolio](#) | [LinkedIn](#)



PROFILE

Computer Science & Engineering undergraduate (CGPA 3.65) with 70+ public repositories and a proven track record of shipping production-grade, full-stack applications across fintech, healthtech, and AI infrastructure. Self-taught systems architect who refactors university projects into scalable enterprise platforms. Built real-world tools for rural healthcare workers, MFS users, and hospital administrators. Open-source enthusiast with a builder's mindset: identify problems, ship solutions, iterate.

CORE COMPETENCIES

Backend	Spring Boot 3, Spring Security 6, JPA/Hibernate, FastAPI, Python, Node.js
Frontend	React 18, Next.js 16, TypeScript, JavaScript, Vite, Thymeleaf, Tailwind CSS
Database	PostgreSQL, SQLite, H2, Supabase, Redis, Flyway, PostGIS
DevOps / Cloud	Docker, GitHub Actions, Cloudflare Workers, Render, Vercel, AWS (EC2, S3)
AI / ML	Ollama, OpenRouter, Whisper, Piper, RAG, LangChain, LLM Integration, Prompt Engineering
System Design	Microservices, Event Sourcing, CQRS, Saga Pattern, Distributed Locking, Observability
Testing	JUnit, Vitest, Playwright, k6 — 300+ tests across projects
Languages	Java, Python, TypeScript, JavaScript, C, C++, Go

TOP PROJECTS

CareBridge RMS — Hospital Management System

Full-stack enterprise hospital platform with real-time queue, bed management, billing, and revenue analytics.

- Built with React + TypeScript + Vite (frontend) and Spring Boot 3 + Spring Security 6 (backend)
- Deployed on Supabase PostgreSQL with Flyway migrations on Render; 2-instance backend, Redis cache + Pub/Sub, Prometheus/Grafana monitoring, RPO 24h / RTO 4h
- Implemented UHID generation with database-locked counters to prevent concurrent duplicates
- Real-time STOMP/WebSocket queue updates for waiting rooms

[GitHub](#)

Nocta — Local AI Chat Interface for Ollama

A single HTML file PWA that turns Ollama into a polished chat app — zero build step, zero dependencies.

- 2,300 lines of vanilla JS in one self-contained HTML file
- Intelligent 4-pass LaTeX preprocessor for KaTeX and GFM table repair for ragged model output
- Offline voice input/output via local Whisper + Piper; installable PWA on Android, iOS, macOS, and desktop
- LAN support to use Ollama from any device on the network; 100+ lines of architecture documentation

[Live Demo](#)

Rentify — Car Rental Platform

Full-stack Spring Boot 3 application with fleet management, booking, AI assistant, and analytics.

- Evolved from a JavaFX university project into a Spring Boot 3 web app
- Dual auth systems: JWT for the mobile REST API and Google OAuth2 for the web dashboard

- Spring Security 6 filter chain with brute-force lockout and idle auto-logout
- OpenRouter AI assistant with fallback models and live DB access to fleet, pricing, rentals, and revenue
- Chart.js dashboards, customer loyalty tiers, PDF receipts via iText, JUnit tests across security & pricing

[GitHub](#)

Rural Health Triage — Bilingual PWA for Community Health Workers

Offline-capable, Bangla/English triage system for rural health workers, built for Bangladesh's 70% rural population.

- React 18 + Vite PWA installable on low-cost Android phones, works fully offline
- FastAPI + Pydantic v2 backend with a Cloudflare Worker edge router for adaptive LLM fallback (free → paid, no redeploy)
- Voice intake in Bangla/English via MediaRecorder → Whisper → translate; prescription OCR with structured medication extraction
- 4-tier triage (Green/Yellow/Red/Black) with differential diagnoses and first-aid guidance; bilingual PDF export
- 46 automated tests (38 backend + 8 frontend) with i18n parity enforced in CI

[GitHub](#)

WalletSync — Multi-Provider MFS Balance Viewer

Personal finance tool for bKash, Nagad, and Rocket users — solves “open three apps to check balance.”

- Next.js 16 + React 18 + TypeScript with a SQLite ledger and pure, injectable-time domain modules for deterministic tests
- Fee calculator using real 2024 BD MFS schedules (1.85% bKash, 1.3% Nagad, 1.8% Rocket)
- Festival forecasting (Eid, Durga Puja, Pohela Boishakh) and O(n) single-pass spending insights per provider
- Recurring-transfer detector using median amount + median gap pattern matching ($\leq 15\%$ variance)
- 302 tests across 30 files; 1000-entry performance under 50ms; production build green

[GitHub](#)

Intelligent Emergency Response & Resource Optimization Platform (Hackathon)

Streaming-first distributed system for nationwide emergency allocation — PSTU National Hackathon 2026 Finalist.

- Go engine with Redis hot state and PostgreSQL/PostGIS durable state
- Costed bipartite matcher for global minimum-cost assignment (ETA, capability, fuel, load)
- mTLS gateway, RBAC, and a tamper-evident audit trail; Saga pattern for allocations
- Network-partition tolerant: local matching with merge on heal; AI severity layer with deterministic fallback (<100ms)
- Scales to 10,000 incidents/minute with partitioned logs and sharded Redis

[GitHub](#)

WORK EXPERIENCE

Open Source Developer (Self-Initiated) | 2025 – Present

- 70+ public repositories spanning full-stack applications, AI tools, and enterprise platforms
- 2,200+ GitHub contributions in the last year — consistent daily shipping
- 50+ live production deployments across the repository portfolio
- Built projects across fintech, healthtech, AI infrastructure, and enterprise domains
- Builds in public on LinkedIn — 4,600+ impressions, international audience

Freelance Full-Stack Developer | 2025 – Present

- Built custom Spring Boot + React solutions for clients (NDA projects)
- Deployed applications on Render, Vercel, Supabase, and Cloudflare Workers
- Implemented OAuth2, JWT, role-based access, and real-time WebSocket features

EDUCATION

Metropolitan University, Bangladesh

B.Sc. in Computer Science & Engineering | 2023 – Present | CGPA: 3.65 / 4.00

ACHIEVEMENTS & RECOGNITION

- Finalist — PSTU National Hackathon 2026 (Intelligent Emergency Response Platform)
- Finalist — Codex Community Hackathon, SUST CSE Carnival 2026
- 2,200+ GitHub contributions in the last year — top tier activity among peers
- 70+ public repositories — one of the most active student developers in Bangladesh

- 4,600+ LinkedIn impressions — content reaching an international software development audience
- Open-source projects featured in developer communities

CERTIFICATIONS & COURSES

- Spring Boot 3 & Spring Security 6 — Hands-on enterprise development
- Full-Stack Web Development — React, Next.js, Node.js
- AI & Machine Learning — Ollama, LLM Integration, RAG Systems
- System Design — Distributed Systems, Microservices, Event Sourcing